挤压激励网络

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摘要—卷积神经网络 （CNN） 的核心构建块是卷积算子，它使网络能够通过在每一层的局部感受野内融合空间和通道信息来构建信息特征。先前的广泛研究已经调查了这种关系的空间分量，试图通过提高整个特征层次结构中的空间编码质量来加强 CNN 的表示能力。在这项工作中，我们专注于通道关系，并提出了一种新的架构单元，我们称之为 “Squeeze-and-Excitation” （SE） 模块，它通过显式建模通道之间的相互依赖关系来自适应地重新校准通道级特征响应。我们表明，这些块可以堆叠在一起形成 SENet 架构，这些架构可以在不同的数据集中非常有效地泛化。我们进一步证明 SE 块为 existin 带来了性能的显着提升

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索引术语 - 挤压和激励、图像表示、注意力、卷积神经网络。

1 引言

C ONVOLUTIONAL 神经网络 （CNN） 已被证明是处理各种视觉任务的有用模型 [1]， [2]， [3]， [4]。在网络的每个卷积层上，一组滤波器表示沿输入通道的邻域空间连接模式——将局部接收场内的空间和通道信息融合在一起。通过将一系列卷积层与非线性激活函数和降采样运算器交错，CNN 能够产生捕获分层模式并获得全局理论感受野的图像表示。计算机视觉研究的一个中心主题是寻找更强大的表示形式，这些表示形式仅捕获图像中对给定任务最突出的那些属性，从而提高性能。作为视觉任务中广泛使用的模型系列，新神经网络架构设计的开发现在代表了这一搜索的关键前沿。最近的研究表明曼斯。进一步的工作试图更好地模拟空间依赖关系 [7]、[8]，并将空间注意力纳入网络结构 [9]。

在本文中，我们研究了网络设计的另一个方面 - 渠道之间的关系。我们引入了一个新的架构单元，我们称之为 *Squeeze-and- Excitation* (SE) 块，其目标是通过显式修改其卷积特征的通道之间的相互依赖关系来提高网络产生的表示质量。为此，我们提出了一种机制，允许网络进行特征重新校准，通过该机制，它可以学习使用全局信息来选择性地强调信息性特征并抑制不太有用的特征。

SE 构建块的结构如图 1 所示。对于将输入 $\mathbf{X}$ 映射到特征映射 $\mathbf{U}$ 的任何给定转换 $\mathbf{F}_{tr}$，其中 $\mathbf{U} \in \mathbb{R}^{H\times W\times C}$，例如卷积，我们可以构造相应的 SE 块来执行特征重新校准。特征 $\mathbf{U}$ 首先通过 *squeeze*作传递，该作通过聚合其空间维度 （$H \times W$） 的特征图来生成通道描述符。这个描述符的功能是生成通道级特征响应的全局分布的嵌入，允许来自网络全局感受野的信息被其所有层使用。聚合之后是 *excitation*作，该作采用简单的自门控机制的形式，该机制将嵌入作为输入并生成每通道调制权重的集合。这些权重应用于特征图 $\mathbf{U}$ 以生成 SE 块的输出，该输出可以直接馈送到网络的后续层。

可以通过简单地堆叠一组 SE 块来构建 SE 网络 （SENet ）。此外，这些 SE 块还可以在网络架构中的一定深度范围内用作原始块的直接替代品。

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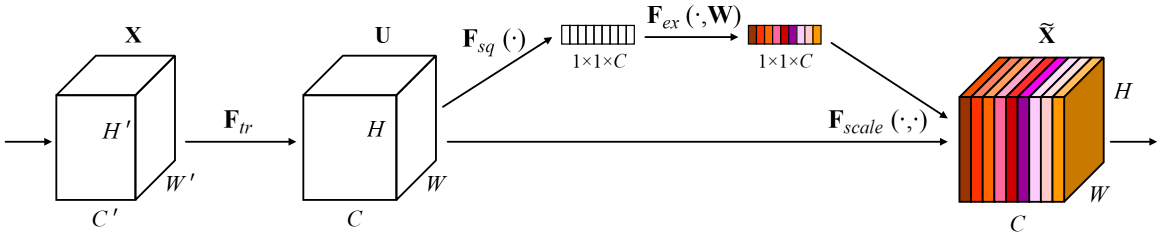


图 1.Squeeze-and-Excitation 模块。

ture （第 6.4 节）。虽然构建基块的模板是通用的，但它在不同深度执行的角色在整个网络中有所不同。在前面的层中，它以与类无关的方式激发信息特征，加强了共享的低级表示。在后面的层中，SE 块变得越来越专业化，并以高度特定于类的方式响应不同的输入（第 7.2 节）。因此，SE 块执行的特征重新校准的好处可以通过网络积累。

新 CNN 架构的设计和开发是一项艰巨的工程任务，通常需要选择许多新的超参数和层配置。相比之下，SE 模块的结构很简单，可以通过用 SE 对应组件替换组件来直接用于现有的最先进的架构，从而可以有效地提高性能。SE 块的计算量也很轻，并且只会略微增加模型复杂性和计算负担。

为了为这些说法提供证据，我们开发了几个 SENet 并对 Ima- geNet 数据集进行了广泛的评估 [10]。我们还展示了 ImageNet 之外的结果，这些结果表明我们方法的好处不仅限于特定的数据集或任务。通过使用 SENet，我们在 ILSVRC 2017 分类竞赛中排名第一。我们最好的模型集成在测试集[1] 上实现了 2.251% 的前 5 个误差。与上一年的获胜者参赛作品（前 5 名误差为 2.991%）相比，这大约有 25% 的相对改进。

2 相关工作

更深层次的架构。VGGNets [11] 和 Inception mod- els [5] 表明，增加网络的深度可以显著提高它能够学习的表征的质量。通过调节每层输入的分布，批量归一化 （BN） [6] 增加了深度网络中学习过程的稳定性，并产生了更平滑的优化表面 [12]。在这些工作的基础上，ResNets 证明，通过使用基于身份的跳跃连接，可以学习更深入、更强大的网络 [13]，[14]。公路网络 [15] 引入了一种门控机制来调节沿捷径连接的信息流。在这些工作之后，对网络层之间的连接进行了进一步的重新表述[16]、[17]、这表明深度网络的学习和表征特性有希望的改进。

另一种但密切相关的研究方向侧重于改进网络中包含的计算元素的功能形式的方法。分组卷积已被证明是增加学习转换基数的流行方法 [18]，[19]。通过多分支卷积 [5]、[6]、[20]、[21] 可以看作是分组算子的自然延伸，可以实现更灵活的算子组合。在以前的工作中，跨通道相关性通常被映射为新的特征组合，即独立于空间结构 [22]、[23] 或通过使用具有 1×1 卷积的标准卷积滤波器 [24] 联合进行映射。这些研究的大部分集中在降低模型和计算复杂性的目标上，反映了一个假设，即通道关系可以被表述为与实例无关的函数与局部感受野的组合。相反，我们声称提供

算法架构搜索。除了上述工作外，还有丰富的研究历史，旨在放弃手动架构设计，而是寻求自动学习网络的结构。该领域的大部分早期工作是在神经进化社区进行的，该社区建立了使用进化方法跨网络拓扑进行搜索的方法[25]，[26]。虽然通常对计算要求很高，但进化搜索已经取得了显著的成功，包括为序列模型[27]、[28]找到好的记忆单元和学习用于大规模图像分类的复杂架构[29]、[30]、[31]。为了减轻这些方法的计算负担，已经提出了基于拉马克继承 [32] 和可微结构搜索 [33] 的方法的有效替代方案。

通过将架构搜索表述为超参数优化，随机搜索 [34] 和其他更诡辩的基于模型的优化技术 [35] 也可用于解决这个问题。拓扑选择作为通过可能设计结构 [37] 的路径，直接架构预测 [38]、[39] 已被提议作为额外的可行架构搜索工具。强化学习技术取得了特别强的结果[40]、[41]、[42]、[43]、[44]。SE 块

1. http://image-net.org/challenges/LSVRC/2017/results

可以用作这些搜索算法的原子构建块，并且已被证明在并发工作中非常有效 [45]。

注意力和门控机制。注意力可以被理解为一种将可用计算资源的分配偏向于信号信息量最大的组成部分的手段 [46]， [47]， [48]， [49]， [50]， [51]。注意力机制已经在许多任务中证明了它们的实用性，包括序列学习[52]、[53]、图像中的定位和理解[9]、[54]、图像描述[55]、[56]和唇读[57]。在这些应用程序中，它可以作为运算符合并到一个或多个层后面，这些层代表更高级别的抽象，以便在模态之间进行调整。一些工作对空间和通道注意力的组合使用提供了有趣的研究 [58]，[59]。Wang等[58]介绍了一种基于沙漏模块[8]的强大树干和掩码注意力机制，该机制插入到深度残差网络的中间阶段之间。相比之下，我们提出的 SE 块包括一个轻量级门控机构，该机构专注于增强 r

3 个挤压和激励模块

Squeeze-and-Excitation 模块是一个计算单元，可以基于将输入 $\mathbf{X} \in \mathbb{R}^{H' \times W' \times C'}$ 映射到特征图 $\mathbf{U} \in \mathbb{R}^{H \times W \times C}$ 的转换 $\mathbf{F}_{tr}$ 构建。在下面的符号中，我们将 $\mathbf{F}_{tr}$ 视为 convo- lutional 运算符，并使用 $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_C]$ 来表示学习的滤波器内核集，其中 $\mathbf{v}_c$ 是指第 c 个滤波器的参数。然后，我们可以将输出写入 $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_C]$，其中

$$\mathbf{u}_c = \mathbf{v}_c * \mathbf{X} = \sum_{s=1}^{C'} \mathbf{v}_c^s * \mathbf{x}^s. \tag{1}$$

这里 $*$ 表示卷积，$\mathbf{v}_c = [\mathbf{v}_c^1, \mathbf{v}_c^2, \ldots, \mathbf{v}_c^{C'}]$、$\mathbf{X} = [\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^{C'}]$ 和 $\mathbf{u}_c \in \mathbb{R}^{H \times W}$。$\mathbf{v}_c^s$ 是一个 2D 空间内核，表示 $\mathbf{v}_c$ 的单个通道，它作用在 $\mathbf{X}$ 的相应通道上。为了简化表示法，省略了偏置项。由于输出是通过所有通道的求和生成的，因此通道依赖关系隐式嵌入在 $\mathbf{v}_c$ 中，但与滤波器捕获的局部空间相关性纠缠在一起。由卷积建模的通道关系本质上是隐式的和局部的（最顶层的通道关系除外）。我们期望通过显式修改通道相互依赖性来增强卷积特征的学习，以便网络能够提高其对信息特征的敏感性，这些信息特征可以被后续转换利用。因此，我们希望为其提供对全局信息的访问权限，并在它们被输入到下一个转换之前，分两步 *squeeze* 和 *excitation* 重新校准滤波器响应

3.1 Squeeze：全局信息嵌入

为了解决利用信道特性的问题，我们首先考虑 output 特征。每个学习到的滤波器都与一个局部感受野一起工作，因此转换输出 $\mathbf{U}$ 的每个单元都无法利用该区域之外的上下文信息。

为了缓解这个问题，我们建议将 *squeeze* 全局空间信息放入通道描述符中。这是通过使用全局平均池生成通道级统计数据来实现的。从形式上讲，统计量 $\mathbf{z} \in \mathbb{R}^C$ 是通过在其空间维度 $H \times W$ 中收缩 $\mathbf{U}$ 来生成的，因此 $\mathbf{z}$ 的第 c 个元素的计算公式为：

$$z_c = \mathbf{F}_{sq}(\mathbf{u}_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} u_c(i, j). \tag{2}$$

Discussion. 转换 $\mathbf{U}$ 的输出可以解释为局部描述符的集合，其统计信息对整个图像具有表现力。利用这些信息在以前的特征工程工作中很普遍[60]、[61]、[62]。我们选择了最简单的聚合技术，即 global average pooling，并指出这里也可以采用更诡辩的策略。

3.2 激励：自适应重新校准

为了利用 *squeeze*作中聚合的信息，我们紧接着进行了第二个作，旨在完全捕获通道依赖性。为了实现这个目标，该函数必须满足两个标准：首先，它必须是灵活的（特别是，它必须能够学习通道之间的非线互），其次，它必须学习非互斥关系，因为我们希望确保允许强调多个通道（而不是强制执行 one-hot 激活）。为了满足这些标准，我们选择采用一种带有 sigmoid 激活的简单门控机制：

$$\mathbf{s} = \mathbf{F}_{ex}(\mathbf{z}, \mathbf{W}) = \sigma(g(\mathbf{z}, \mathbf{W})) = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z})), \tag{3}$$

其中 δ 是指 ReLU [63] 函数，$\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ 和 $\mathbf{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ 是指 ReLU [63] 函数。为了限制模型复杂性并帮助泛化，我们通过在非线性周围形成两个全连接 （FC） 层的瓶颈来参数化门控机制，即具有缩减比的降维层 r (这个参数选择在第 6.1 节)中讨论，一个 ReLU，然后是一个增维层返回到转换输出 $\mathbf{U}$ 的通道维度。区块的最终输出是通过使用 activations $\mathbf{s}$ 重新缩放 $\mathbf{U}$ 来获得的：

$$\widetilde{\mathbf{x}}_c = \mathbf{F}_{scale}(\mathbf{u}_c, s_c) = s_c \, \mathbf{u}_c, \tag{4}$$

其中 $\widetilde{\mathbf{X}} = [\widetilde{\mathbf{x}}_1, \widetilde{\mathbf{x}}_2, \ldots, \widetilde{\mathbf{x}}_C]$ 和 $\mathbf{F}_{scale}(\mathbf{u}_c, s_c)$ 是指标量 s_c 和特征图 $\mathbf{u}_c \in \mathbb{R}^{H \times W}$ 之间的通道乘法。

Discussion. 激励运算符将特定于输入的描述符 $\mathbf{z}$ 映射到一组通道权重。在这方面，SE 块本质上引入了对输入的动态条件，这可以被视为通道上的自注意函数，这些通道的关系不限于卷积滤波器响应的局部感受野。

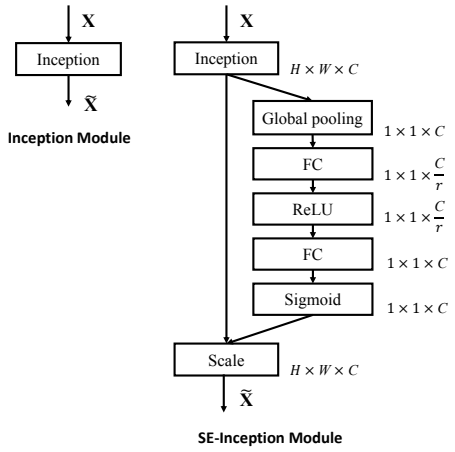


图 2.原始 Inception 模块（左）和 SE- Inception 模块（右）的架构。

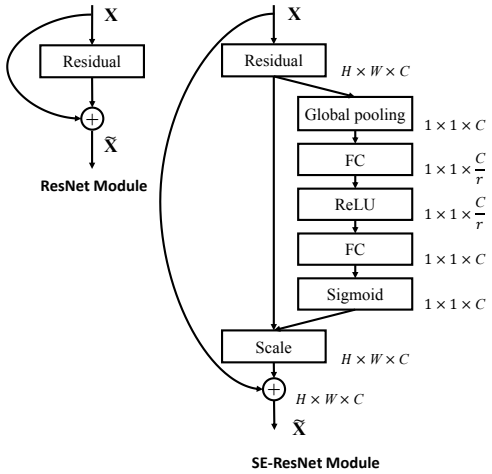


图 3.原始 Residual 模块（左）和 SE-ResNet 模块（右）的架构。

3.3 实例化

SE 块可以集成到 VGGNet [11] 等标准架构中，方法是在每次卷积后的非线性之后插入。此外，SE 块的灵活性意味着它可以直接应用于标准卷积之外的转换。为了说明这一点，我们通过将 SE 块合并到几个更复杂的架构示例中来开发 SENet，如下所述。

我们首先考虑构建集成网络的 SE 块 [5]。在这里，我们简单地将转换 $\mathbf{F}_{tr}$ 视为一个完整的 Inception 模块（见图 2），通过对架构中的每个此类模块进行此更改，我们获得了一个 *SE-Inception* 网络。SE 块也可以直接与残差网络一起使用（图 3 描述了 *SE-ResNet* 模块的架构）。在这里，SE 块转换 $\mathbf{F}_{tr}$ 被视为残差模块的非标识分支。*Squeeze* 和 *Excitation* 都在与身份分支求和之前起作用。将 SE 块与 ResNeXt [19]、Inception-ResNet [21]、MobileNet [64] 和 ShuffleNet [65] 集成的更多变体可以通过遵循类似的方案来构建。有关 SENet 架构的具体示例，表 1 中给出了 SE-ResNet-50 和 SE-ResNeXt-50 的详细说明。

SE 块的灵活性的一个结果是，有几种可行的方法可以将其集成到这些架构中。因此，为了评估对用于将 SE 块合并到网络架构中的集成策略的敏感性，我们还在第 6.5 节中提供了消融实验，探索了块包含的不同设计。

4 模型和计算复杂性

要使提议的 SE 模块设计具有实际用途，它必须在提高性能和增加模型复杂性之间提供良好的权衡。为了说明与该模块相关的计算负担，我们以 ResNet-50 和 SE-ResNet-50 之间的比较为例。对于 224×224 像素的输入图像，ResNet-50 需要在单个正向传递中使用 ∼3.86 GFLOP。每个 SE 区块都使用 *squeeze* 阶段和 *excitation* 阶段的两个小 FC 层，然后是廉价的通道缩放作。总的来说，当将第 3.2) 节中引入的缩减率 r（设置为 16 时，SE-ResNet-50 需要 ∼3.87 GFLOPs，相当于比原始 ResNet-50 相对增加 0.26%。作为这种轻微的额外计算负担的交换，SE-ResNet-50 的精度超过了 ResNet-50，实际上接近了需要 ∼7.58 GFLOP 的更深 ResNet-101 网络的精度（表 2）。

实际上，通过 ResNet-50 向前和向后进行一次传递需要 190 毫秒，而 SE-ResNet-50 需要 209 毫秒，训练小批量包含 256 张图像（两个计时都是在具有 8 个 NVIDIA Titan X GPU 的服务器上执行的）。我们认为这代表了合理的运行时开销，随着全局池和小型内积作在流行的 GPU 库中得到进一步优化，这种开销可能会进一步减少。由于它对嵌入式设备应用的重要性，我们进一步对每个模型的 CPU 推理时间进行了基准测试：对于 224×224 像素的输入图像，ResNet-50 需要 164 毫秒，而 SE-ResNet-50 需要 167 毫秒。我们认为，SE 块产生的少量额外计算成本是合理的，因为它对模型性能的贡献是合理的。

接下来，我们考虑拟议的 SE 区块引入的额外参数。这些附加参数仅由门控机制的两个 FC 层产生，因此只占总网络容量的一小部分。具体来说，这些 FC 层的权重参数引入的总数由下式给出：

$$\frac{2}{r}\sum_{s=1}^{S} N_s \cdot {C_s}^2, \tag{5}$$

其中 r 表示缩减率，S 是指阶段数（阶段是指在公共空间维度的特征图上运行的块的集合），C_s 表示输出通道的维度，N_s 表示阶段 s (的重复块数}当在 FC 层中使用偏置项时， 引入的参数和计算成本通常可以忽略不计)。SE-ResNet-50 引入了 ∼2.500 万个附加参数，超越了

表 1（左）ResNet-50 [13]。（中）SE-ResNet-50 的。（右）带有 32×4d 模板的 SE-ResNeXt-50。剩余构建块的形状和具有特定参数设置的作列在括号内，阶段中堆叠块的数量显示在外部。*fc* 后面的内括号表示 SE 模块中两个全连接层的输出维度。

Output size	ResNet-50	SE-ResNet-50	SE-ResNeXt-50 (32×4d)
112×112	conv, 7×7, 64, stride 2		
56×56	max pool, 3×3, stride 2		
	$\begin{bmatrix} \text{conv}, 1 \times 1, 64 \\ \text{conv}, 3 \times 3, 64 \\ \text{conv}, 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} \text{conv}, 1 \times 1, 64 \\ \text{conv}, 3 \times 3, 64 \\ \text{conv}, 1 \times 1, 256 \\ fc, [16, 256] \end{bmatrix} \times 3$	$\begin{bmatrix} \text{conv}, 1 \times 1, 128 \\ \text{conv}, 3 \times 3, 128 \quad C = 32 \\ \text{conv}, 1 \times 1, 256 \\ fc, [16, 256] \end{bmatrix} \times 3$
28×28	$\begin{bmatrix} \text{conv}, 1 \times 1, 128 \\ \text{conv}, 3 \times 3, 128 \\ \text{conv}, 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} \text{conv}, 1 \times 1, 128 \\ \text{conv}, 3 \times 3, 128 \\ \text{conv}, 1 \times 1, 512 \\ fc, [32, 512] \end{bmatrix} \times 4$	$\begin{bmatrix} \text{conv}, 1 \times 1, 256 \\ \text{conv}, 3 \times 3, 256 \quad C = 32 \\ \text{conv}, 1 \times 1, 512 \\ fc, [32, 512] \end{bmatrix} \times 4$
14×14	$\begin{bmatrix} \text{conv}, 1 \times 1, 256 \\ \text{conv}, 3 \times 3, 256 \\ \text{conv}, 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} \text{conv}, 1 \times 1, 256 \\ \text{conv}, 3 \times 3, 256 \\ \text{conv}, 1 \times 1, 1024 \\ fc, [64, 1024] \end{bmatrix} \times 6$	$\begin{bmatrix} \text{conv}, 1 \times 1, 512 \\ \text{conv}, 3 \times 3, 512 \quad C = 32 \\ \text{conv}, 1 \times 1, 1024 \\ fc, [64, 1024] \end{bmatrix} \times 6$
7×7	$\begin{bmatrix} \text{conv}, 1 \times 1, 512 \\ \text{conv}, 3 \times 3, 512 \\ \text{conv}, 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} \text{conv}, 1 \times 1, 512 \\ \text{conv}, 3 \times 3, 512 \\ \text{conv}, 1 \times 1, 2048 \\ fc, [128, 2048] \end{bmatrix} \times 3$	$\begin{bmatrix} \text{conv}, 1 \times 1, 1024 \\ \text{conv}, 3 \times 3, 1024 \quad C = 32 \\ \text{conv}, 1 \times 1, 2048 \\ fc, [128, 2048] \end{bmatrix} \times 3$
1×1	global average pool, 1000-d fc, softmax		

表 2 ImageNet 验证集的单作物错误率 （%） 和复杂性比较。*original* 列是指原始论文中报告的结果（ResNets 的结果从网站获得：https://github.com/Kaiminghe/deep-residual-networks）。为了实现公平的比较，我们重新训练基线模型并在 *re-implementation* 列中报告分数。*SENet* 列是指添加了 SE 块的相应架构。括号中的数字表示与重新实施的基准相比的性能改进。† 表示模型已在验证集的非黑名单子集上进行评估（这在 [21] 中有更详细的讨论），这可能会略微改善结果。VGG-16 和 SE-VGG-16 使用批量归一化进行训练。

	original		re-implementation			SENet		
	top-1 err.	top-5 err.	top-1 err.	top-5 err.	GFLOPs	top-1 err.	top-5 err.	GFLOPs
ResNet-50 [13]	24.7	7.8	24.80	7.48	3.86	$23.29_{(1.51)}$	$6.62_{(0.86)}$	3.87
ResNet-101 [13]	23.6	7.1	23.17	6.52	7.58	$22.38_{(0.79)}$	$6.07_{(0.45)}$	7.60
ResNet-152 [13]	23.0	6.7	22.42	6.34	11.30	$21.57_{(0.85)}$	$5.73_{(0.61)}$	11.32
ResNeXt-50 [19]	22.2	-	22.11	5.90	4.24	$21.10_{(1.01)}$	$5.49_{(0.41)}$	4.25
ResNeXt-101 [19]	21.2	5.6	21.18	5.57	7.99	$20.70_{(0.48)}$	$5.01_{(0.56)}$	8.00
VGG-16 [11]	-	-	27.02	8.81	15.47	$25.22_{(1.80)}$	$7.70_{(1.11)}$	15.48
BN-Inception [6]	25.2	7.82	25.38	7.89	2.03	$24.23_{(1.15)}$	$7.14_{(0.75)}$	2.04
Inception-ResNet-v2 [21]	$19.9^{\dagger}$	$4.9^{\dagger}$	20.37	5.21	11.75	$19.80_{(0.57)}$	$4.79_{(0.42)}$	11.76

∼ResNet-50 需要 2500 万个参数，相当于 ∼10% 的增长。在实践中，这些参数中的大部分来自网络的最后阶段，其中激励作在最多数量的通道上执行。然而，我们发现，这个相对昂贵的 SE 块的最后阶段可以以很小的性能成本（<0.1% 在 ImageNet 上排名前 5 的错误）被移除，将相对参数增加减少到 ∼4%，这在参数使用是关键考虑因素的情况下可能很有用（参见第 6.4 节和第 7.2 节进一步讨论）。

5 实验

在本节中，我们进行了实验，以研究 SE 块在一系列任务、数据集和模型架构中的有效性。

5.1 图像分类

为了评估 SE 块的影响，我们首先在 ImageNet 2012 数据集 [10] 上进行实验，该数据集包括 1.2800 万张训练图像和 50K 验证来自 1000 个不同类别的图像。我们在训练集上训练网络，并在验证集上报告 top-1 和 top-5 错误。

每个基线网络架构及其相应的 SE 对应物都使用相同的优化方案进行训练。我们遵循标准做法，使用比例和比例 [5] 通过随机裁剪进行数据增强，大小为 224×224 像素（或 Inception-ResNet-v2 [21] 和 SE-Inception-ResNet-v2 为 299×299），并执行随机水平翻转。每个输入图像都通过平均 RGB 通道减法进行归一化。所有模型都在我们的分布式学习系统 *ROCS* 上进行训练，该系统旨在处理大型网络的高效并行训练。使用动量为 0.9 且小批量大小为 1024 的同步 SGD 进行优化。初始学习率设置为 0.6，每 30 个 epoch 降低 10 倍。使用 [66] 中描述的权重初始化策略，从头开始训练模型 100 个 epoch。默认情况下，第 3.2) 节中的缩减比率 r (设置为 16（除非另有说明）。

在评估模型时，我们应用中心裁剪，以便从每张图像中裁剪 224×224 像素，然后

TABLE 3
Single-crop error rates (%) on the ImageNet validation set and complexity comparisons. MobileNet refers to "1.0 MobileNet-224" in [64] and ShuffleNet refers to "ShuffleNet $1 \times (g = 3)$" in [65]. The numbers in brackets denote the performance improvement over the re-implementation.

	original		re-implementation				SENet			
	top-1 err.	top-5 err.	top-1 err.	top-5 err.	MFLOPs	Params	top-1 err.	top-5 err.	MFLOPs	Params
MobileNet [64]	29.4	-	28.4	9.4	569	4.2M	$25.3_{(3.1)}$	$7.7_{(1.7)}$	572	4.7M
ShuffleNet [65]	32.6	-	32.6	12.5	140	1.8M	$31.0_{(1.6)}$	$11.1_{(1.4)}$	142	2.4M

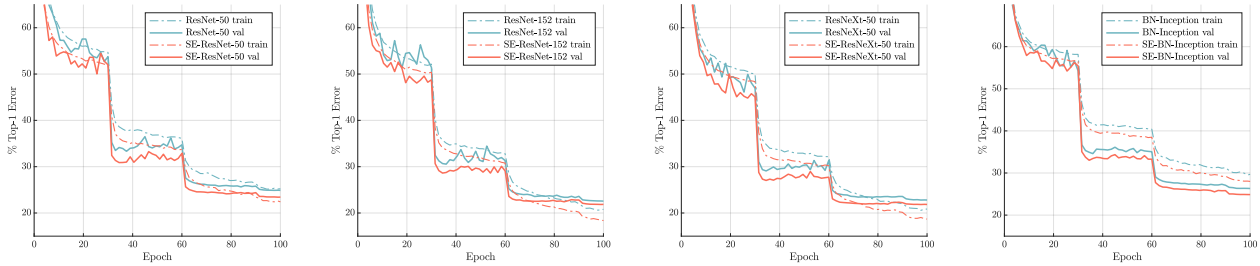


Fig. 4. Training baseline architectures and their SENet counterparts on ImageNet. SENets exhibit improved optimisation characteristics and produce consistent gains in performance which are sustained throughout the training process.

its shorter edge is first resized to 256 (299×299 from each image whose shorter edge is first resized to 352 for Inception-ResNet-v2 and SE-Inception-ResNet-v2).

Network depth. We begin by comparing SE-ResNet against ResNet architectures with different depths and report the results in Table 2. We observe that SE blocks consistently improve performance across different depths with an extremely small increase in computational complexity. Remarkably, SE-ResNet-50 achieves a single-crop top-5 validation error of 6.62%, exceeding ResNet-50 (7.48%) by 0.86% and approaching the performance achieved by the much deeper ResNet-101 network (6.52% top-5 error) with only half of the total computational burden (3.87 GFLOPs vs. 7.58 GFLOPs). This pattern is repeated at greater depth, where SE-ResNet-101 (6.07% top-5 error) not only matches, but outperforms the deeper ResNet-152 network (6.34% top-5 error) by 0.27%. While it should be noted that the SE blocks themselves add depth, they do so in an extremely computationally efficient manner and yield good returns even at the point at which extending the depth of the base architecture achieves diminishing returns. Moreover, we see that the gains are consistent across a range of different network depths, suggesting that the improvements induced by SE blocks may be complementary to those obtained by simply increasing the depth of the base architecture.

Integration with modern architectures. We next study the effect of integrating SE blocks with two further state-of-the-art architectures, Inception-ResNet-v2 [21] and ResNeXt (using the setting of 32×4d) [19], both of which introduce additional computational building blocks into the base network. We construct SENet equivalents of these networks, SE-Inception-ResNet-v2 and SE-ResNeXt (the configuration of SE-ResNeXt-50 is given in Table 1) and report results in Table 2. As with the previous experiments, we observe significant performance improvements induced by the introduction of SE blocks into both architectures. In particular, SE-ResNeXt-50 has a top-5 error of 5.49% which is superior to both its direct counterpart ResNeXt-50 (5.90% top-5 error) as well as the deeper ResNeXt-101 (5.57% top-5 error), a model which has almost twice the total number of parameters and computational overhead. We note a slight difference in performance between our re-implementation of Inception-ResNet-v2 and the result reported in [21]. However, we observe a similar trend with regard to the effect of SE blocks, finding that SE counterpart (4.79% top-5 error) outperforms our reimplemented Inception-ResNet-v2 baseline (5.21% top-5 error) by 0.42% as well as the reported result in [21].

We also assess the effect of SE blocks when operating on *non-residual* networks by conducting experiments with the VGG-16 [11] and BN-Inception architecture [6]. To facilitate the training of VGG-16 from scratch, we add Batch Normalization layers after each convolution. We use identical training schemes for both VGG-16 and SE-VGG-16. The results of the comparison are shown in Table 2. Similarly to the results reported for the residual baseline architectures, we observe that SE blocks bring improvements in performance on the non-residual settings.

To provide some insight into influence of SE blocks on the optimisation of these models, example training curves for runs of the baseline architectures and their respective SE counterparts are depicted in Fig. 4. We observe that SE blocks yield a steady improvement throughout the optimisation procedure. Moreover, this trend is fairly consistent across a range of network architectures considered as baselines.

Mobile setting. Finally, we consider two representative architectures from the class of mobile-optimised networks, MobileNet [64] and ShuffleNet [65]. For these experiments, we used a minibatch size of 256 and slightly less aggressive data augmentation and regularisation as in [65]. We trained the models across 8 GPUs using SGD with momentum (set to 0.9) and an initial learning rate of 0.1 which was reduced by a factor of 10 each time the validation loss plateaued. The total training process required ~ 400 epochs (enabling us to reproduce the baseline performance of [65]). The results reported in Table 3 show that SE blocks consistently improve

TABLE 4
Classification error (%) on CIFAR-10.

	original	SENet
ResNet-110 [14]	6.37	5.21
ResNet-164 [14]	5.46	4.39
WRN-16-8 [67]	4.27	3.88
Shake-Shake 26 2x96d [68] + Cutout [69]	2.56	2.12

TABLE 5
Classification error (%) on CIFAR-100.

	original	SENet
ResNet-110 [14]	26.88	23.85
ResNet-164 [14]	24.33	21.31
WRN-16-8 [67]	20.43	19.14
Shake-Even 29 2x4x64d [68] + Cutout [69]	15.85	15.41

TABLE 6
Single-crop error rates (%) on Places365 validation set.

	top-1 err.	top-5 err.
Places-365-CNN [72]	41.07	11.48
ResNet-152 (ours)	41.15	11.61
SE-ResNet-152	**40.37**	**11.01**

TABLE 7
Faster R-CNN object detection results (%) on COCO *minival* set.

	AP@IoU=0.5	AP
ResNet-50	57.9	38.0
SE-ResNet-50	61.0	40.4
ResNet-101	60.1	39.9
SE-ResNet-101	62.7	41.9

the accuracy by a large margin at a minimal increase in computational cost.

Additional datasets. We next investigate whether the benefits of SE blocks generalise to datasets beyond ImageNet. We perform experiments with several popular baseline architectures and techniques (ResNet-110 [14], ResNet-164 [14], WideResNet-16-8 [67], Shake-Shake [68] and Cutout [69]) on the CIFAR-10 and CIFAR-100 datasets [70]. These comprise a collection of 50k training and 10k test 32×32 pixel RGB images, labelled with 10 and 100 classes respectively. The integration of SE blocks into these networks follows the same approach that was described in Section 3.3. Each baseline and its SENet counterpart are trained with standard data augmentation strategies [24], [71]. During training, images are randomly horizontally flipped and zero-padded on each side with four pixels before taking a random 32×32 crop. Mean and standard deviation normalisation is also applied. The setting of the training hyperparameters (e.g. minibatch size, initial learning rate, weight decay) match those suggested by the original papers. We report the performance of each baseline and its SENet counterpart on CIFAR-10 in Table 4 and performance on CIFAR-100 in Table 5. We observe that in every comparison SENets outperform the baseline architectures, suggesting that the benefits of SE blocks are not confined to the ImageNet dataset.

5.2 Scene Classification

We also conduct experiments on the Places365-Challenge dataset [73] for scene classification. This dataset comprises 8 million training images and $36,500$ validation images across 365 categories. Relative to classification, the task of scene understanding offers an alternative assessment of a model's ability to generalise well and handle abstraction. This is because it often requires the model to handle more complex data associations and to be robust to a greater level of appearance variation.

We opted to use ResNet-152 as a strong baseline to assess the effectiveness of SE blocks and follow the training and evaluation protocols described in [72], [74]. In these experiments, models are trained from scratch. We report the results in Table 6, comparing also with prior work. We observe that SE-ResNet-152 (11.01% top-5 error) achieves a lower validation error than ResNet-152 (11.61% top-5 error), providing evidence that SE blocks can also yield improvements for scene classification. This SENet surpasses the previous state-of-the-art model Places-365-CNN [72] which has a top-5 error of 11.48% on this task.

5.3 Object Detection on COCO

We further assess the generalisation of SE blocks on the task of object detection using the COCO dataset [75]. As in previous work [19], we use the *minival* protocol, i.e., training the models on the union of the 80k training set and a 35k val subset and evaluating on the remaining 5k val subset. Weights are initialised by the parameters of the model trained on the ImageNet dataset. We use the Faster R-CNN [4] detection framework as the basis for evaluating our models and follow the hyperparameter setting described in [76] (i.e., end-to-end training with the '2x' learning schedule). Our goal is to evaluate the effect of replacing the trunk architecture (ResNet) in the object detector with SE-ResNet, so that any changes in performance can be attributed to better representations. Table 7 reports the validation set performance of the object detector using ResNet-50, ResNet-101 and their SE counterparts as trunk architectures. SE-ResNet-50 outperforms ResNet-50 by 2.4% (a relative 6.3% improvement) on COCO's standard AP metric and by 3.1% on AP@IoU=0.5. SE blocks also benefit the deeper ResNet-101 architecture achieving a 2.0% improvement (5.0% relative improvement) on the AP metric. In summary, this set of experiments demonstrate the generalisability of SE blocks. The induced improvements can be realised across a broad range of architectures, tasks and datasets.

5.4 ILSVRC 2017 Classification Competition

SENets formed the foundation of our submission to the ILSVRC competition where we achieved first place. Our winning entry comprised a small ensemble of SENets that employed a standard multi-scale and multi-crop fusion strategy to obtain a top-5 error of 2.251% on the test set. As part of this submission, we constructed an additional model, *SENet-154*, by integrating SE blocks with a modified ResNeXt [19] (the details of the architecture are provided in Appendix). We compare this model with prior work on the ImageNet validation set in Table 8 using standard crop

TABLE 8
Single-crop error rates (%) of state-of-the-art CNNs on ImageNet validation set with crop sizes 224×224 and 320×320 / 299×299.

	224×224		320×320 / 299×299	
	top-1 err.	top-5 err.	top-1 err.	top-5 err.
ResNet-152 [13]	23.0	6.7	21.3	5.5
ResNet-200 [14]	21.7	5.8	20.1	4.8
Inception-v3 [20]	-	-	21.2	5.6
Inception-v4 [21]	-	-	20.0	5.0
Inception-ResNet-v2 [21]	-	-	19.9	4.9
ResNeXt-101 (64×4d) [19]	20.4	5.3	19.1	4.4
DenseNet-264 [17]	22.15	6.12	-	-
Attention-92 [58]	-	-	19.5	4.8
PyramidNet-200 [77]	20.1	5.4	19.2	4.7
DPN-131 [16]	19.93	5.12	18.55	4.16
SENet-154	**18.68**	**4.47**	**17.28**	**3.79**

TABLE 9
Comparison (%) with state-of-the-art CNNs on ImageNet validation set using larger crop sizes/additional training data. †This model was trained with a crop size of 320×320.

	extra data	crop size	top-1 err.	top-5 err.
Very Deep PolyNet [78]	-	331	18.71	4.25
NASNet-A (6 @ 4032) [42]	-	331	17.3	3.8
PNASNet-5 (N=4,F=216) [35]	-	331	17.1	3.8
SENet-154†	-	320	16.88	3.58
AmoebaNet-C [79]	-	331	16.5	3.5
ResNeXt-101 32×48d [80]	✓	224	14.6	2.4

sizes (224×224 and 320×320). We observe that SENet-154 achieves a top-1 error of 18.68% and a top-5 error of 4.47% using a 224×224 centre crop evaluation, which represents the strongest reported result.

Following the challenge there has been a great deal of further progress on the ImageNet benchmark. For comparison, we include the strongest results that we are currently aware of in Table 9. The best performance using only ImageNet data was recently reported by [79]. This method uses reinforcement learning to develop new policies for data augmentation during training to improve the performance of the architecture searched by [31]. The best overall performance was reported by [80] using a ResNeXt-101 $32 \times 48d$ architecture. This was achieved by pretraining their model on approximately one billion weakly labelled images and finetuning on ImageNet. The improvements yielded by more sophisticated data augmentation [79] and extensive pretraining [80] may be complementary to our proposed changes to the network architecture.

6 Ablation Study

In this section we conduct ablation experiments to gain a better understanding of the effect of using different configurations on components of the SE blocks. All ablation experiments are performed on the ImageNet dataset on a single machine (with 8 GPUs). ResNet-50 is used as the backbone architecture. We found empirically that on ResNet architectures, removing the biases of the FC layers in the excitation operation facilitates the modelling of channel dependencies, and use this configuration in the following experiments. The data augmentation strategy follows the approach described in Section 5.1. To allow us to study the upper limit of performance for each variant, the learning rate is initialised to 0.1 and training continues until the validation loss plateaus[2] ($\sim$300 epochs in total). The learning rate is then reduced by a factor of 10 and then this process is repeated (three times in total). Label-smoothing regularisation [20] is used during training.

TABLE 10
Single-crop error rates (%) on ImageNet and parameter sizes for SE-ResNet-50 at different reduction ratios. Here, *original* refers to ResNet-50.

Ratio r	top-1 err.	top-5 err.	Params
2	22.29	6.00	45.7M
4	22.25	6.09	35.7M
8	22.26	5.99	30.7M
16	22.28	6.03	28.1M
32	22.72	6.20	26.9M
original	23.30	6.55	25.6M

6.1 Reduction ratio

The reduction ratio r introduced in Eqn. 5 is a hyperparameter which allows us to vary the capacity and computational cost of the SE blocks in the network. To investigate the trade-off between performance and computational cost mediated by this hyperparameter, we conduct experiments with SE-ResNet-50 for a range of different r values. The comparison in Table 10 shows that performance is robust to a range of reduction ratios. Increased complexity does not improve performance monotonically while a smaller ratio dramatically increases the parameter size of the model. Setting $r = 16$ achieves a good balance between accuracy and complexity. In practice, using an identical ratio throughout a network may not be optimal (due to the distinct roles performed by different layers), so further improvements may be achievable by tuning the ratios to meet the needs of a given base architecture.

6.2 Squeeze Operator

We examine the significance of using global average pooling as opposed to global max pooling as our choice of squeeze operator (since this worked well, we did not consider more sophisticated alternatives). The results are reported in Table 11. While both max and average pooling are effective, average pooling achieves slightly better performance, justifying its selection as the basis of the squeeze operation. However, we note that the performance of SE blocks is fairly robust to the choice of specific aggregation operator.

6.3 Excitation Operator

We next assess the choice of non-linearity for the excitation mechanism. We consider two further options: ReLU and tanh, and experiment with replacing the sigmoid with these

2. For reference, training with a 270 epoch fixed schedule (reducing the learning rate at 125, 200 and 250 epochs) achieves top-1 and top-5 error rates for ResNet-50 and SE-ResNet-50 of (23.21%, 6.53%) and (22.20%, 6.00%) respectively.

TABLE 11
Effect of using different squeeze operators in SE-ResNet-50 on ImageNet (error rates %).

Squeeze	top-1 err.	top-5 err.
Max	22.57	6.09
Avg	**22.28**	**6.03**

TABLE 12
Effect of using different non-linearities for the excitation operator in SE-ResNet-50 on ImageNet (error rates %).

Excitation	top-1 err.	top-5 err.
ReLU	23.47	6.98
Tanh	23.00	6.38
Sigmoid	**22.28**	**6.03**

alternative non-linearities. The results are reported in Table 12. We see that exchanging the sigmoid for tanh slightly worsens performance, while using ReLU is dramatically worse and in fact causes the performance of SE-ResNet-50 to drop below that of the ResNet-50 baseline. This suggests that for the SE block to be effective, careful construction of the excitation operator is important.

6.4 Different stages

We explore the influence of SE blocks at different stages by integrating SE blocks into ResNet-50, one stage at a time. Specifically, we add SE blocks to the intermediate stages: stage_2, stage_3 and stage_4, and report the results in Table 13. We observe that SE blocks bring performance benefits when introduced at each of these stages of the architecture. Moreover, the gains induced by SE blocks at different stages are complementary, in the sense that they can be combined effectively to further bolster network performance.

6.5 Integration strategy

Finally, we perform an ablation study to assess the influence of the location of the SE block when integrating it into existing architectures. In addition to the proposed SE design, we consider three variants: (1) SE-PRE block, in which the SE block is moved before the residual unit; (2) SE-POST block, in which the SE unit is moved after the summation with the identity branch (after ReLU) and (3) SE-Identity block, in which the SE unit is placed on the identity connection in parallel to the residual unit. These variants are illustrated in Figure 5 and the performance of each variant is reported in Table 14. We observe that the SE-PRE, SE-Identity and proposed SE block each perform similarly well, while usage

TABLE 13
Effect of integrating SE blocks with ResNet-50 at different stages on ImageNet (error rates %).

Stage	top-1 err.	top-5 err.	GFLOPs	Params
ResNet-50	23.30	6.55	3.86	25.6M
SE_Stage_2	23.03	6.48	3.86	25.6M
SE_Stage_3	23.04	6.32	3.86	25.7M
SE_Stage_4	22.68	6.22	3.86	26.4M
SE_All	22.28	6.03	3.87	28.1M

TABLE 14
Effect of different SE block integration strategies with ResNet-50 on ImageNet (error rates %).

Design	top-1 err.	top-5 err.
SE	22.28	6.03
SE-PRE	22.23	6.00
SE-POST	22.78	6.35
SE-Identity	22.20	6.15

TABLE 15
Effect of integrating SE blocks at the 3x3 convolutional layer of each residual branch in ResNet-50 on ImageNet (error rates %).

Design	top-1 err.	top-5 err.	GFLOPs	Params
SE	22.28	6.03	3.87	28.1M
SE_3×3	22.48	6.02	3.86	25.8M

of the SE-POST block leads to a drop in performance. This experiment suggests that the performance improvements produced by SE units are fairly robust to their location, provided that they are applied prior to branch aggregation.

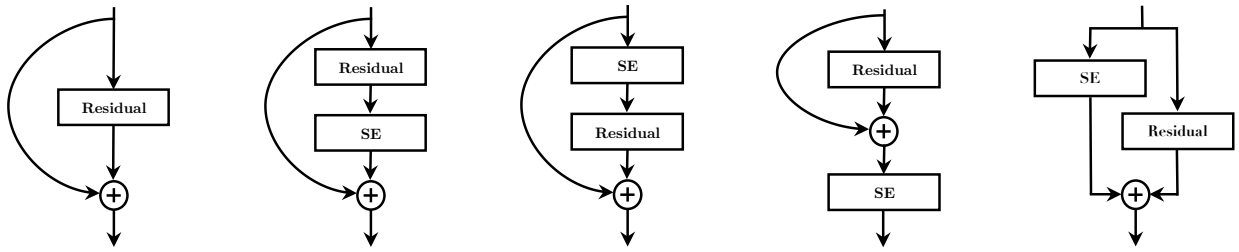
In the experiments above, each SE block was placed outside the structure of a residual unit. We also construct a variant of the design which moves the SE block inside the residual unit, placing it directly after the 3×3 convolutional layer. Since the 3×3 convolutional layer possesses fewer channels, the number of parameters introduced by the corresponding SE block is also reduced. The comparison in Table 15 shows that the SE_3×3 variant achieves comparable classification accuracy with fewer parameters than the standard SE block. Although it is beyond the scope of this work, we anticipate that further efficiency gains will be achievable by tailoring SE block usage for specific architectures.

7 Role of SE blocks

Although the proposed SE block has been shown to improve network performance on multiple visual tasks, we would also like to understand the relative importance of the squeeze operation and how the excitation mechanism operates in practice. A rigorous theoretical analysis of the representations learned by deep neural networks remains challenging, we therefore take an empirical approach to examining the role played by the SE block with the goal of attaining at least a primitive understanding of its practical function.

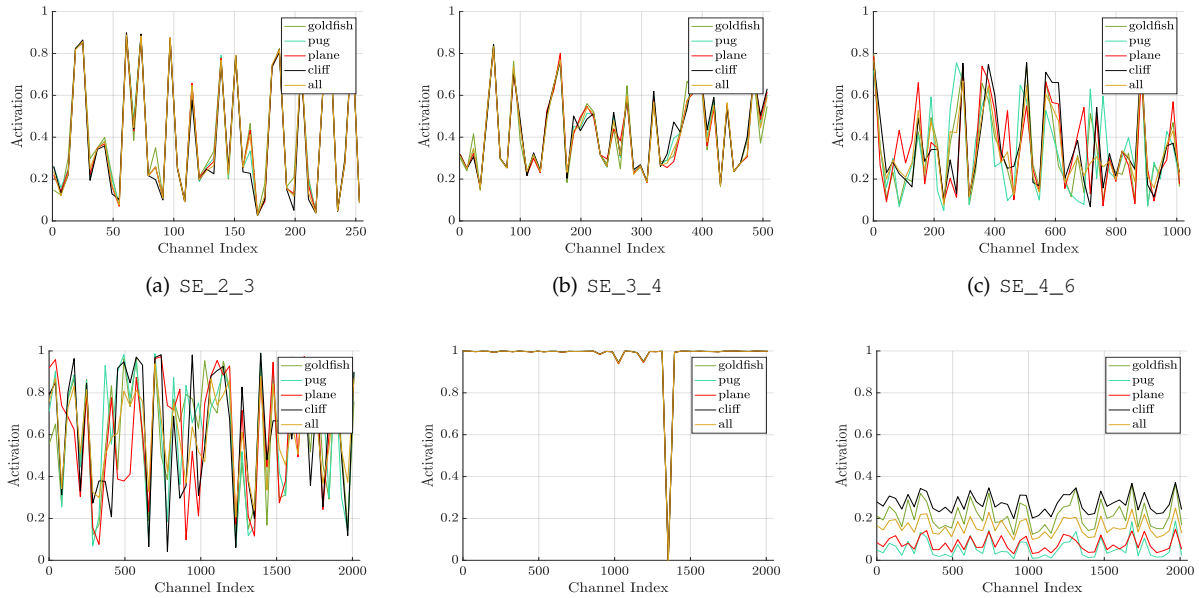
7.1 Effect of Squeeze

To assess whether the global embedding produced by the squeeze operation plays an important role in performance, we experiment with a variant of the SE block that adds an equal number of parameters, but does not perform global average pooling. Specifically, we remove the pooling operation and replace the two FC layers with corresponding 1×1 convolutions with identical channel dimensions in the excitation operator, namely *NoSqueeze*, where the excitation output maintains the spatial dimensions as input. In contrast to the SE block, these point-wise convolutions can only remap the channels as a function of the output of a local operator. While in practice, the later layers of a deep network will typically possess a (theoretical) global



(a) Residual block (b) Standard SE block (c) SE-PRE block (d) SE-POST block (e) SE-Identity block

Fig. 5. SE block integration designs explored in the ablation study.



(a) SE_2_3 (b) SE_3_4 (c) SE_4_6

(d) SE_5_1 (e) SE_5_2 (f) SE_5_3

Fig. 6. Activations induced by the Excitation operator at different depths in the SE-ResNet-50 on ImageNet. Each set of activations is named according to the following scheme: SE_stageID_blockID. With the exception of the unusual behaviour at SE_5_2, the activations become increasingly class-specific with increasing depth.

TABLE 16
Effect of Squeeze operator on ImageNet (error rates %).

	top-1 err.	top-5 err.	GFLOPs	Params
ResNet-50	23.30	6.55	3.86	25.6M
NoSqueeze	22.93	6.39	4.27	28.1M
SE	**22.28**	**6.03**	3.87	28.1M

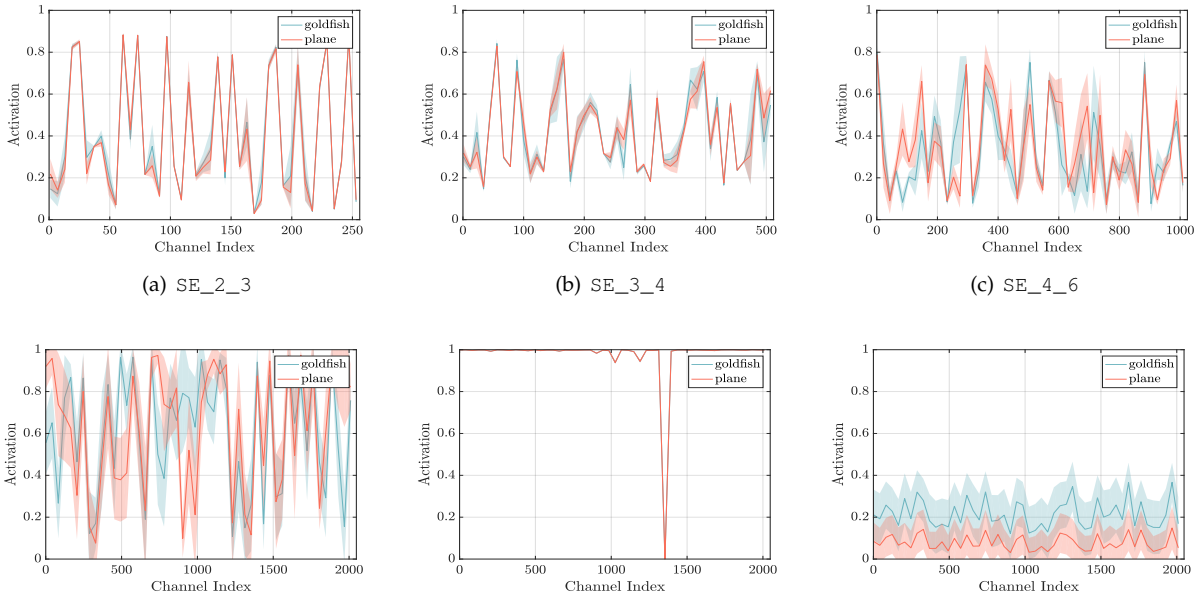
receptive field, global embeddings are no longer directly accessible throughout the network in the *NoSqueeze* variant. The accuracy and computational complexity of both models are compared to a standard ResNet-50 model in Table 16. We observe that the use of global information has a significant influence on the model performance, underlining the importance of the squeeze operation. Moreover, in comparison to the *NoSqueeze* design, the SE block allows this global information to be used in a computationally parsimonious manner.

7.2 Role of Excitation

To provide a clearer picture of the function of the excitation operator in SE blocks, in this section we study example activations from the SE-ResNet-50 model and examine their distribution with respect to different classes and different input images at various depths in the network. In particular, we would like to understand how excitations vary across images of different classes, and across images within a class.

We first consider the distribution of excitations for different classes. Specifically, we sample four classes from the ImageNet dataset that exhibit semantic and appearance diversity, namely *goldfish*, *pug*, *plane* and *cliff* (example images from these classes are shown in Appendix). We then draw fifty samples for each class from the validation set and compute the average activations for fifty uniformly sampled channels in the last SE block of each stage (immediately prior to downsampling) and plot their distribution in Fig. 6. For reference, we also plot the distribution of the mean activations across all of the 1000 classes.

We make the following three observations about the role of the *excitation* operation. First, the distribution across different classes is very similar at the earlier layers of the network, e.g. SE_2_3. This suggests that the importance of feature channels is likely to be shared by different classes in the early stages. The second observation is that at greater



(a) SE_2_3 (b) SE_3_4 (c) SE_4_6

(d) SE_5_1 (e) SE_5_2 (f) SE_5_3

Fig. 7. Activations induced by *Excitation* in the different modules of SE-ResNet-50 on image samples from the goldfish and plane classes of ImageNet. The module is named "SE_stageID_blockID".

depth, the value of each channel becomes much more class-specific as different classes exhibit different preferences to the discriminative value of features, e.g. SE_4_6 and SE_5_1. These observations are consistent with findings in previous work [81], [82], namely that earlier layer features are typically more general (e.g. class agnostic in the context of the classification task) while later layer features exhibit greater levels of specificity [83].

Next, we observe a somewhat different phenomena in the last stage of the network. SE_5_2 exhibits an interesting tendency towards a saturated state in which most of the activations are close to one. At the point at which all activations take the value one, an SE block reduces to the identity operator. At the end of the network in the SE_5_3 (which is immediately followed by global pooling prior before classifiers), a similar pattern emerges over different classes, up to a modest change in scale (which could be tuned by the classifiers). This suggests that SE_5_2 and SE_5_3 are less important than previous blocks in providing recalibration to the network. This finding is consistent with the result of the empirical investigation in Section 4 which demonstrated that the additional parameter count could be significantly reduced by removing the SE blocks for the last stage with only a marginal loss of performance.

Finally, we show the mean and standard deviations of the activations for image instances within the same class for two sample classes (*goldfish* and *plane*) in Fig. 7. We observe a trend consistent with the inter-class visualisation, indicating that the dynamic behaviour of SE blocks varies over both classes and instances within a class. Particularly in the later layers of the network where there is considerable diversity of representation within a single class, the network learns to take advantage of feature recalibration to improve its discriminative performance [84]. In summary, SE blocks produce instance-specific responses which nevertheless function to support the increasingly class-specific needs of the model at different layers in the architecture.

8 Conclusion

In this paper we proposed the SE block, an architectural unit designed to improve the representational power of a network by enabling it to perform dynamic channel-wise feature recalibration. A wide range of experiments show the effectiveness of SENets, which achieve state-of-the-art performance across multiple datasets and tasks. In addition, SE blocks shed some light on the inability of previous architectures to adequately model channel-wise feature dependencies. We hope this insight may prove useful for other tasks requiring strong discriminative features. Finally, the feature importance values produced by SE blocks may be of use for other tasks such as network pruning for model compression.

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Appendix: Details of SENet-154

SENet-154 is constructed by incorporating SE blocks into a modified version of the 64×4d ResNeXt-152 which extends

(a) goldfish (b) pug (c) plane (d) cliff

Fig. 8. Sample images from the four classes of ImageNet used in the experiments described in Sec. 7.2.

the original ResNeXt-101 [19] by adopting the block stacking strategy of ResNet-152 [13]. Further differences to the design and training of this model (beyond the use of SE blocks) are as follows: (a) The number of the first 1×1 convolutional channels for each bottleneck building block was halved to reduce the computational cost of the model with a minimal decrease in performance. (b) The first 7×7 convolutional layer was replaced with three consecutive 3×3 convolutional layers. (c) The 1×1 down-sampling projection with stride-2 convolution was replaced with a 3×3 stride-2 convolution to preserve information. (d) A dropout layer (with a dropout ratio of 0.2) was inserted before the classification layer to reduce overfitting. (e) Label-smoothing regularisation (as introduced in [20]) was used during training. (f) The parameters of all BN layers were frozen for the last few training epochs to ensure consistency between training and testing. (g) Training was performed with 8 servers (64 GPUs) in parallel to enable large batch sizes (2048). The initial learning rate was set to 1.0.

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